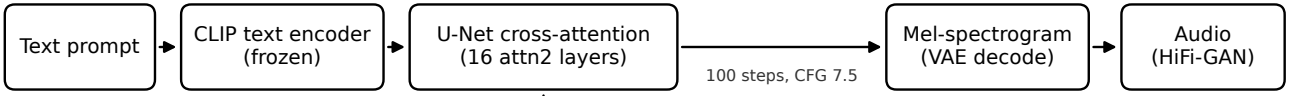
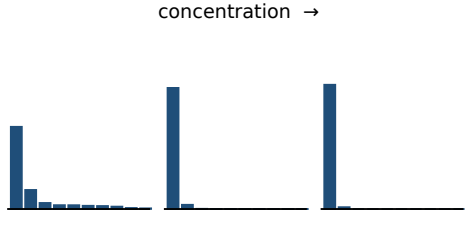


Cross-attention intervention ($Z = QK^\top/\sqrt{d}$, output $= AV$)

- Dense (baseline) $A_{\text{dense}} = \text{Softmax}(Z)$
- Sharp Softmax $A_{\text{sharp}} = \text{Softmax}(\beta Z)$
- Entmax (sparse) $A_{\text{sparse}} = \text{Entmax}_{1.5}(Z)$
- PLADIS $A = A_{\text{dense}} + \lambda_{\text{entmax}}\,(A_{\text{sparse}} - A_{\text{dense}})$

16 text cross-attention layers only; self-attention, scheduler and CFG unchanged

Conceptual (not measured): attention over text tokens